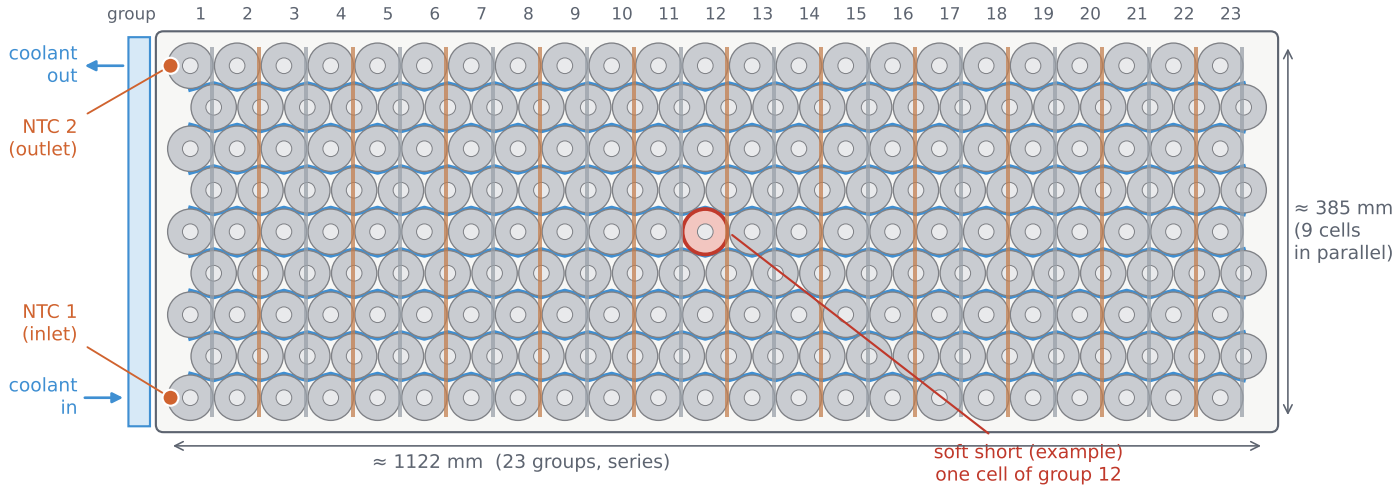
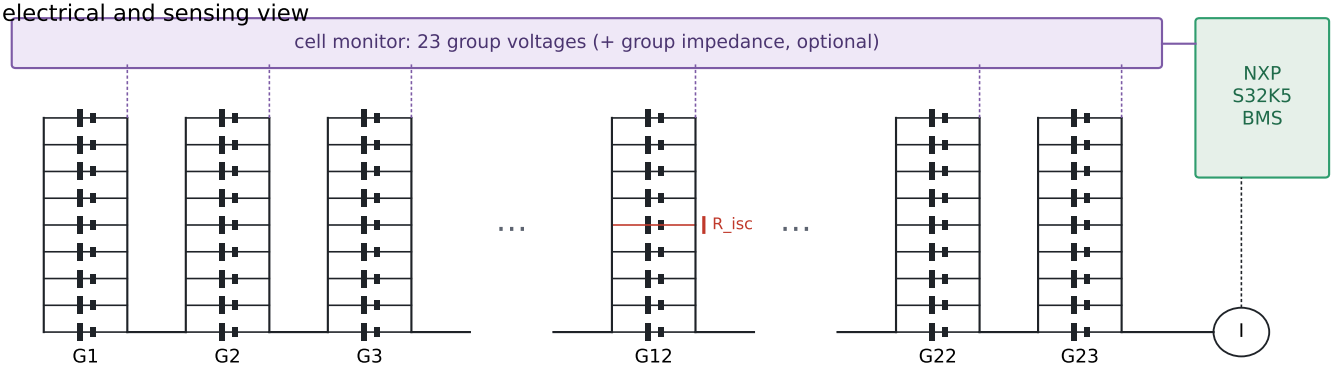
a top view to scale: 207 cells, 23s9p, staggered, cooling ribbons between rows



b electrical and sensing view



- 2 NTCs (coolant inlet, outlet) → S32K5

A short in one cell of a group is fed by the whole group: the group voltage drifts against the other 22 groups at rest, by about 1/9 of what a single cell would show.

Reference module, not a specific product: 23s9p as one of four modules of the Tesla Model Y 4680 pack (92s9p, 828 cells); BYD FC4680 cells (∅46.3 × 80.2 mm). Sensing as assumed in this work. Ribbon, busbar and manifold geometry schematic; cell pitch and packing to scale.